

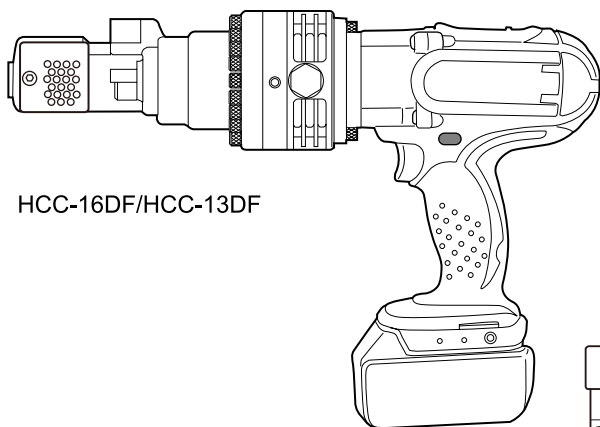
Ogura® ELECTRO-HYDRAULIC CORDLESS REBAR CUTTER

Model:HCC-13DF

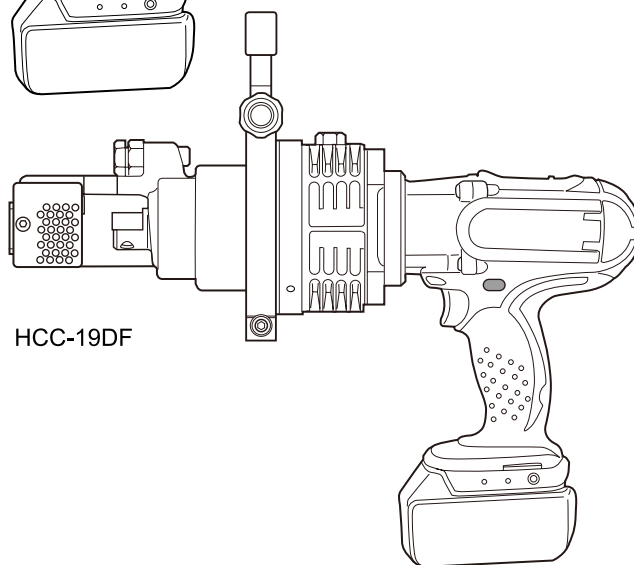
Model:HCC-16DF

Model:HCC-19DF

Instruction Manual



HCC-16DF/HCC-13DF



HCC-19DF

Before installing and operating this machine, read, understand and follow all instructions and operating procedures. Keep these instructions with the machine.

Read, understand and follow all safety instructions and operating procedures. If you do not understand the instructions, or if conditions are not correct for proper operation, DO NOT OPERATE THE MACHINE. Consult your supervisor or other responsible person.

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WARNING



Read and understand this instruction manual before operating this machine. Failure to follow operating instructions could result in death or serious injury.

WARNING LABELS RELATED TO SAFETY



Flying debris and loud noise hazards. Wear ear and eye protection.



Moving blade. Keep hands clear while machine is operating. Turn power off before servicing.

SAVE THESE INSTRUCTIONS

Meaning of "caution" and "warning" indications

Caution: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. This is also used to alert against unsafe practices associated with events that could lead to personal injury.

Warning: Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.

***** Ogura & Co., Ltd. shall not be responsible for any incidental damages or personal injuries resulting from negligence of Warnings and Safety Instructions contained in the Instruction Manual.**

Important Safety Instructions

- The "important safety instructions" described below should always be followed to reduce the risk of a fire, electric shock, and personal injury.
- Before operating the machine, read and carefully follow these instructions.
- This manual should be retained for future reference.

WARNING

1. Before use, read this Instruction Manual thoroughly.

2. Use only the charger and battery specified in this Instruction Manual.

- Any combination, except that specified in this manual, may cause injury or damage due to explosion.

3. Charge properly.

- Use this charger only with the rated AC power supply. Do not use booster transformers or vehicle generator, or use the charger to supply DC power. Incorrect use may cause overheating and fire.
- Do not charge the battery at a temperature under 10°C(50°F) or more than 40°C (104°F) as this may cause an explosion or fire.
- Do not charge the battery on paper, cardboard, cloth, cushions, mats, carpets, vinyl etc..
- Charge the battery in a well ventilated place. Do not cover the battery or charger during charging as this may cause an explosion or fire.
- If during charging there is an abnormal generation of heat, stop the operation immediately by disconnecting the power supply.
- Overheating may cause smoke, fire, and explosion.
- Disconnect the power when the charger is not in use to prevent any possibility of electric shock or fire.

4. Do not short-circuit the pins of the battery or place metal objects close to it, as this may cause overheating and explosion due to short circuit.

5. Guard against an electric shock.

- Do not touch the power plug with a wet or damp hands as this may result in an electric shock.

6. Consider the work area environment.

- Do not expose the charger and battery to rain or use them in damp or wet locations, as this may cause overheating or electric shock.
- Keep the work area well lighted. Working where

there is insufficient light may cause an accident.

- Do not use or charge the battery where there are flammable fluids or gases, as this may cause explosion or fire.
- Check for the presence of persons on the ground during operation in an elevated place.
- Any material or tools dropped may cause an accident.

7. Wear Safety Glasses and Protective Clothing

- Always wear safety glasses, safety footwear, safety gloves, and any other mandated or necessary protective clothing while using this equipment. Failure to do so may result in injury.

8. Secure the work.

- Use clamps or a vice to hold the work. This frees both hands to properly hold, control, and operate the tool. Failure to properly secure the work may result in injury.

9. Disconnect the tools power supply, by removing the Battery and engaging the Trigger Switch Lock, whenever one of the following situations occur.

- The tool is not in use or is being serviced.
 - Any parts, such as a blade, are being replaced.
 - There is a recognised hazard.
- Failure to do so may result in accidental operation and damage or injury.

10. Avoid accidental operation.

- Do not carry the tool by the Trigger Switch as this may cause accidental operation and damage or injury.

11. Only use the specified accessories or attachments.

- Use only the accessories or attachments described in this Instruction Manual and the Ogura catalog. Use of any other accessories or attachments may result in an accident or injury.

12. Do not expose the battery to fire as it may explode or generate poisonous substances.

Important Safety Instructions

CAUTION

1. Hold the tool securely while it is in use.

- If the tool is not held securely, you may be injured.

2. Keep your hands and face away from the moving parts.

- They may cause an injury.

3. Release the Trigger Switch immediately to stop operation when the tool is out of order or makes an abnormal sound during use. Have it inspected and repaired by an authorized service center.

- Failure to do so may result in damage or injury.

4. If you drop or strike the tool, check carefully that the body is not damaged, cracked, or deformed.

- Any such damage could cause injury.

5. The battery cover should be fitted at all times when the battery is not in use.

6. The ORC-19DF/ORC-16DF is an electro-hydraulic tool. The oil reservoir was filled before delivery. Do not add oil unless the machine operates abnormally.

7. Use only Ogura original parts.

8. Metal cutting blades have sharp edges. Handle them carefully to avoid being cut.

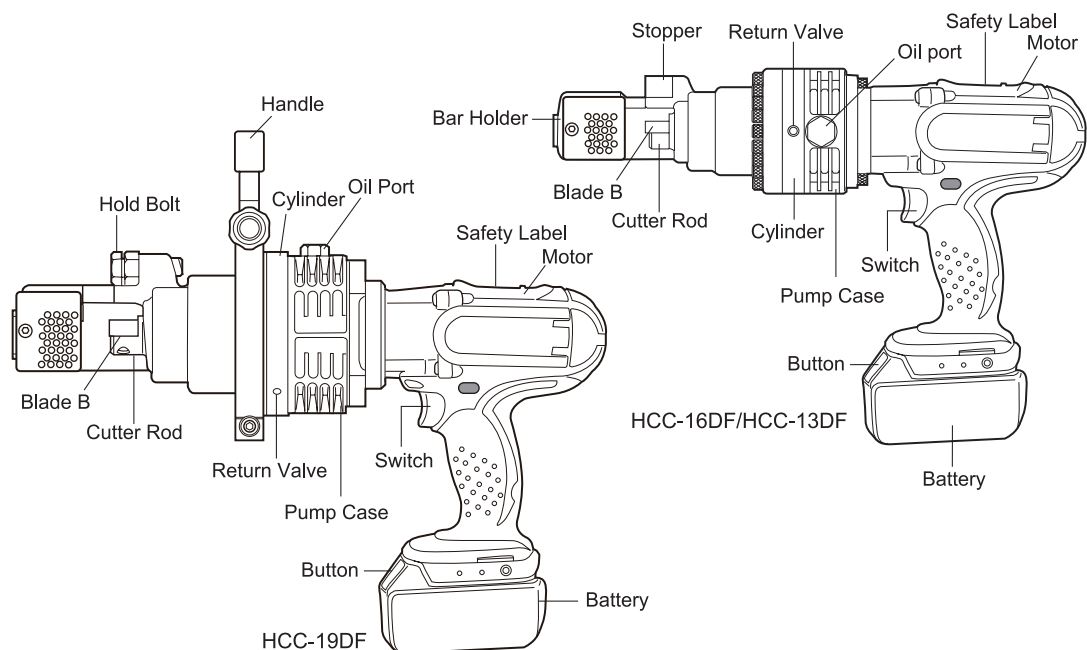
9. Damaged, deformed or cracked blades may cause serious accidents as well as impair operation. Replace with new genuine blades immediately.

10. When using an extension cord with the charger, it is recommended to use a cord with the cross section below and with a length as short as possible. For charging outdoors, use a cord rated for outdoor use.

Relation between usable cord thickness and maximum length

Cord size (Nominal cross sectional area of conductor)	Maximum cord length
0.75 mm ²	20 m
1.25 mm ²	30 m

NAME OF THE PARTS



Specifications

Model		HCC-13DF	HCC-16DF	HCC-19DF
Motor		DC magnet motor		
Battery		Lithium-ion battery		
Voltage		18V D.C.		
Max. Cutting Capacity (Dia.mm)		13 mm / 1/2"	16 mm / 5/8"	20 mm / 3/4"
Grade 40: Tensile Strength 490 N / mm ² 70,000 PSI				
Grade 60: Tensile Strength 620 N / mm ² 90,000 PSI				
Cutting Speed		2.5 seconds	4.0 seconds	6.5 seconds
Dimensions (L x W x H mm)		331 x 262 x 99	380 x 265 x 108	412 x 289 x 234 (including a holding handle)
Weight (including a battery)		4.7 kg	6.5 kg	10.5 kg
Sound Noise	Sound Pressure Level	80.8 dB(A) with no load		
	Sound Power Level	91.8 dB(A) with no load		
Vibration (EN 60745-2-8:2009)	Total value	Less than 2.5 m/s ²		
	Uncertainty	1.5 m/s ²		
Battery Charger		DC18RC T		
Input Voltage		110-120V / 220-240 V, A.C. in single phase		
Input Frequency		50-60 Hz		
Input Capacity		240 W		
Output Voltage		7.2~18V D.C.		
Output Current		9A D.C.		

● Specifications and design may be subject to change without prior notice.

■ Standard accessories (HCC-16DF)

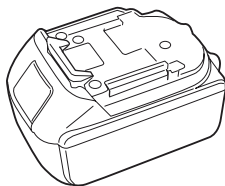
Battery charger	-----	1
Battery	-----	1
Battery cover	-----	1
Hex wrenches (M3,M4,M5,M6)	----- One each	
Double-end Spanner (14 mm X 17 mm)	-----	1
Hydraulic oil #46 (35 cm ³)	-----	1
Carrying case	-----	1

■ Standard accessories (HCC-13DF)

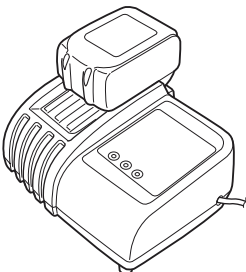
Battery charger	-----	1
Battery	-----	1
Battery cover	-----	1
Hex wrenches (M3,M4,M5)	----- One each	
Double-end Spanner (14 mm X 17 mm)	-----	1
Hydraulic oil #46 (35 cm ³)	-----	1
Carrying case	-----	1

■ Standard accessories (HCC-19DF)

Battery charger	-----	1
Battery	-----	1
Battery cover	-----	1
Hex wrenches (M3,M4,M5,M6)	----- One each	
Double-end Spanner (17 mm X 19 mm)	-----	1
Hydraulic oil #46 (150 cm ³)	-----	1
Carrying case	-----	1

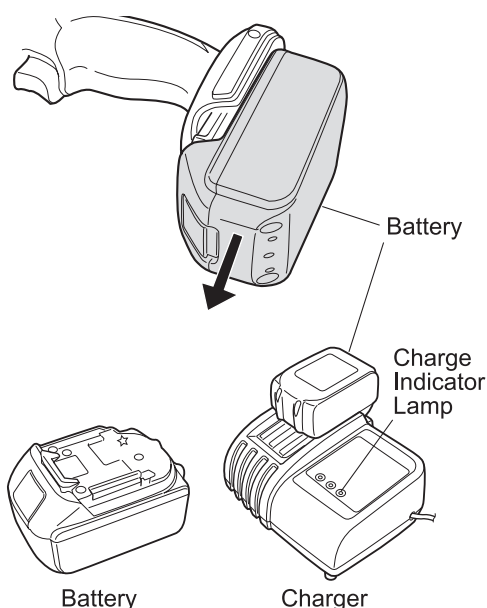


Battery



Charger

BATTERY CHARGING PROCEDURE



Before using the tool the Battery should be charged using the Charger provided.

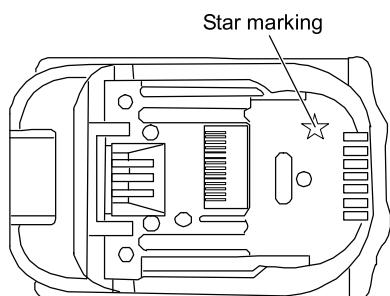
1. To remove the Battery first depress the two locking buttons (one on either side) and then holding the battery firmly pull it out of the housing.
 2. Connect the Battery Charger to an AC power supply of 110-120V/220-240V. The green Indicator Lamp will blink.
 3. Fully insert the Battery into the Battery Charger making sure that the lug on the Battery is inline with the slot in the Battery Charger. The charge Indicator Lamp (red) will illuminate to show charging has started.
 4. When charging is completed the charge Indicator Lamp will turn green and the Electronic Buzzer will beep for about five seconds.
 5. Remove the Battery from the Battery Charger and replace it in the tool using the reverse procedure to its removal.
- For the correct combination of Battery and Battery Charger and the charging time, see the table on the left.
 - After charging is completed, install the battery in the tool making sure it is locked in position.

Notes

- Battery Charger DC18RC T is only to be used for the charging the batteries. Do not use it for any other purpose.
- A new Battery or a Battery that has not used for a long time may not be fully charged by one time charging. In such cases full charge should be achieved after the Battery has been discharged and charged two or three times.
- The charge Indicator Lamp may repeatedly blink (red), when the Battery is put on charge immediately after being used or after being exposed to direct sunlight for a long time. In such cases allow the Battery to cool down before putting on charge.
- When the Battery has reached the end of its useful life, or the Battery/Battery Charger contacts are dirty, the charge Indicator Lamp will repeatedly blink (red and green) by turn and the Electronic Buzzer will beep for about 20 seconds after the start of charging.
- The charging time may be lengthened if the battery is continuously charged.

Battery protection system (Lithium-ion battery with star marking)

Lithium-ion batteries with a star marking are equipped with a protection system. This system automatically cuts off power to the tool to extend battery life. The tool will automatically stop during operation if the tool and/or battery are placed under one of the following conditions:



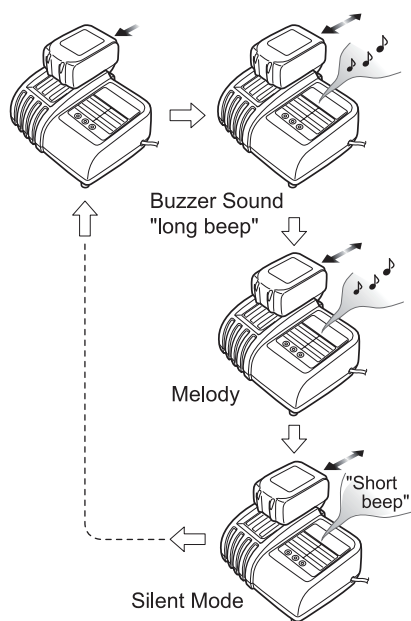
Overloaded:

The tool is operated in a manner that causes it to draw an abnormally high current. In this situation, release the trigger switch on the tool and stop the application that caused the tool to become overloaded. Then pull the trigger switch again to restart. If the tool does not start, the battery is overheated. In this situation, let the battery cool before pulling the trigger switch again.

Low battery voltage:

The remaining battery capacity is too low and the tool will not operate. In this situation, remove and recharge the battery.

Changing the Preset Melody for “Charging completed”



1. Inserting the battery cartridge into charger brings out last preset brief melody sound of completed charging.
2. Removing and re-inserting it within five seconds after this action makes the melody sound change.
3. Every time removing and re-inserting it within another five seconds after this, the melody sound changes in order.
4. When the desired melody sound comes out, leave the battery cartridge being inserted and the charge will begin. When a "short beep" mode is selected, no completed charging signals come out. (Silent mode)
5. With finish of charge, the green light remains lit with the red light going out and the melody sound preset at the insertion of battery cartridge or buzzer sound (a long beep) comes out to notify completion of charge. (In selected silent mode, no sounds come out.)
6. Preset melody sound remains stored even when the charger is unplugged.

Trickle charge (Maintenance charge)

If you leave the battery cartridge in the charger to prevent spontaneous discharging after full charge, the charger will switch into its "trickle charge (maintenance charge)" mode and keep the battery cartridge fresh and fully charged. To maintain product SAFETY and RELIABILITY, repairs, any other maintenance or adjustment should be performed by OGURA Authorized Service Centers, always using OGURA replacement parts.

Cooling system

- This charger is equipped with cooling fan heated battery in order to enable the battery to prove its own performance. Sound of cooling air comes out during, which means no trouble on the charger.
- Yellow light will flash for warning in the following cases.
 - Trouble on cooling fan
 - Incomplete cool down of battery, such as, being clogged with dustThe battery can be charged in spite of the yellow warning light. But the charging time will be longer than usual in this case.
Check the sound of cooling fan, vent on the charger and battery, which can be sometime clogged with dust.
- The cooling system is in order although no sound of cooling fan comes out, if the yellow warning light will not flash.
- Always keep clean the vent on charger and battery for cooling.
- The products should be sent to repair or maintenance, if the yellow warning light will frequently flash.

BATTERY CHARGING

Conditioning charge

Conditioning charge can extend the life of battery by automatically searching the optimum charging condition for the batteries in every situation. The battery employed in the following conditions repeatedly requires "conditioning charge" to prevent fast wear out. In that case, yellow light lights up.

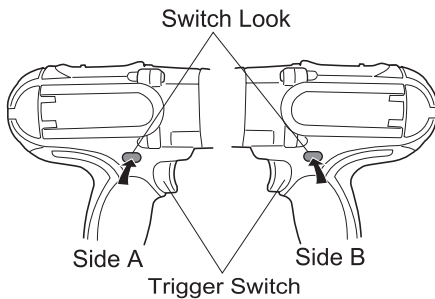
1. Recharge of battery with its high temperature
2. Recharge of battery with its low temperature
3. Recharge of full charged battery
4. Over-discharge of battery (continue to discharge battery in spite of down of power.)

The charging time of such battery is longer than usual.

CAUTION

1. Do not short battery.
2. Do not destroy battery by fire.
3. Do not expose battery to water or rain.
4. **CAUTION** - To reduce risk of injury, charge only OGURA specified batteries. Other types of batteries may burst causing personal injury and damage.
5. Non-rechargeable batteries cannot be charged in this battery charger.
6. Use a power source with the voltage specified on the nameplate of the charger.
7. Do not charge the battery in presence of flammable liquids or gases.
8. Do not expose charger to rain or snow.
9. Never carry charger by cord or yank it to disconnect from receptacle.
10. After charging or before attempting any maintenance or cleaning, unplug the charger from the power source. Pull by plug rather than cord whenever disconnecting charger.
11. Make sure cord is located where it will not be stepped on, tripped over, or otherwise subjected to damage or stress.
12. Do not operate charger with damaged cord or plug - replace them immediately.
13. Do not operate or disassemble charger if it has received a sharp blow, been dropped, or otherwise damaged in any way; take it to a qualified serviceman. Incorrect use or reassembly may result in a risk of electric shock or fire.
14. The battery charger is not intended for use by young children or infirm persons without supervision.
15. Young children should be supervised to ensure that they do not play with the battery charger.
16. Do not charge battery when temperature is BELOW 10°C (50°F) or ABOVE 40°C (104°F).
17. Do not attempt to use a step-up transformer, an engine generator or DC power receptacle.
18. Do not allow anything to cover or clog the charger vents.

OPERATING PROCEDURE



Read, understand and follow all safety instructions and operating procedures. If you do not understand the instructions, or if conditions are not correct for proper operation, do not operate this tool. Consult your supervisor or other responsible person.

⚠ WARNING

Before the Battery is inserted into the tool, pull and release the Trigger Switch to ensure that it returns when released.

- The motor is on when the Trigger Switch is pulled and off when the Trigger Switch is released.

Switch Lock Operation

- Push in the Switch Lock on side A. The Switch is un-locked and the Trigger can be operated.
- Push in the Switch Lock on side B. The Switch is locked and the Trigger cannot be operated.

⚠ CAUTION

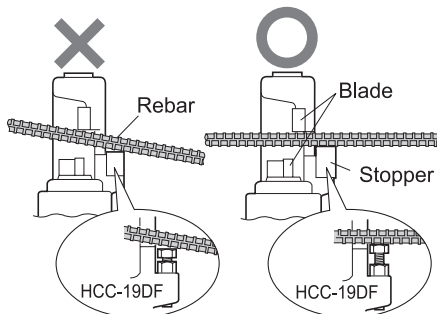
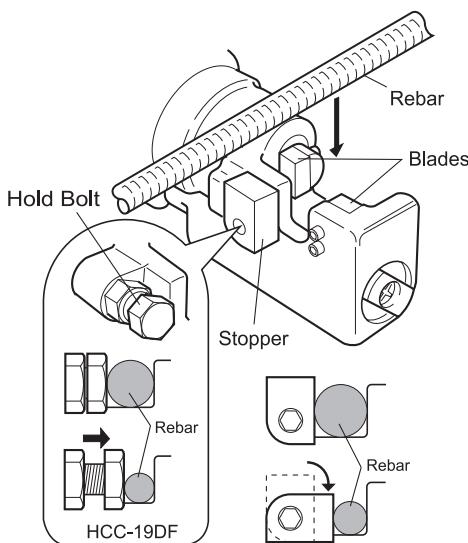
The Trigger Switch should be locked at all times when not in use.

⚠ WARNING

1. Before operation, confirm that the position of the operator, relative to the tool, and the surrounding area is safe for operation. Put on safety glasses and wear protective clothing.
2. Refer to the tool specifications in this manual and do not cut rebar of size or hardness that exceeds the cutting capacity of the tool.
3. Do not cut material other than rebar. Please ask the manufacturer if you want to cut other materials.

⚠ WARNING

Replace damaged (chipped, broken, cracked) or deformed blades immediately. The blade will not cut true and may fracture or break causing serious personal injury.



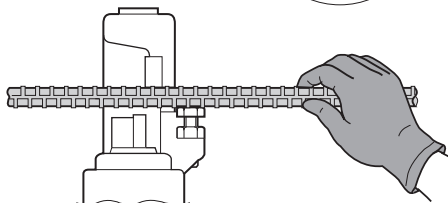
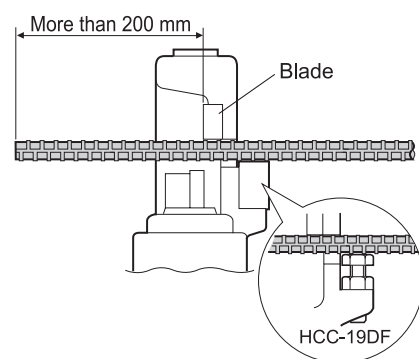
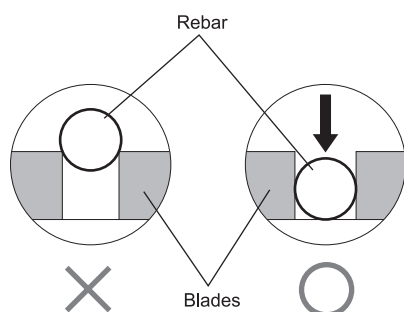
Cutting Procedure

1. Position the rebar to be cut between the blades.

Adjust the Stopper (HCC-13DF/16DF) or the Hold Bolt (HCC-19DF) according to the diameter of the rebar to be cut so that the rebar is at 90 degrees to the blades as shown in the figure on the left. The Stopper (HCC-13DF/16DF) or the Hold Bolt (HCC-19DF) supports the rebar and keeps it perpendicular to the blades when cutting.

WARNING

When cutting rebar, adjust the Stopper (HCC-13DF/16DF) or the Hold Bolt (HCC-19DF) according to the diameter of there bar to be cut so that the rebar is at 90 degrees to the blades as shown in the figure on the left. Without this adjustment, the cut piece may fly off and cause serious injury to the operator or bystanders. Never fail to check the position of the operator relative to the tool and confirm the safety of the operator and surrounding area.



2. Position the rebar to be cut so that it is fully between the blades as shown in the figure on the left.

WARNING

If the rebar to be cut is not positioned fully between the blades, the blades will be damaged; the rebar will be ejected violently and may cause serious personal injury.

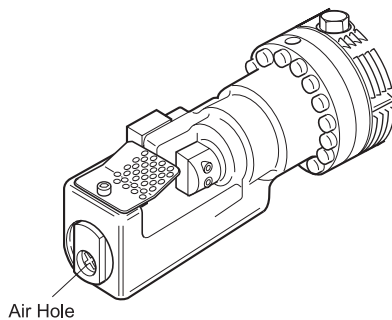
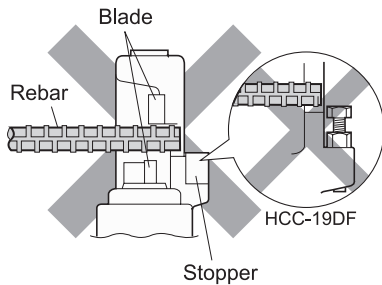
WARNING

Do not cut rebar when the piece to be cut off is less than 200 mm in length. Cutting shorter length may cause the rebar to fly off during cut and may result in serious personal injury.

WARNING

Do not cut rebar when it is not properly supported by the Stopper (HCC-13DF/16DF) or the Hold Bolt (HCC-19DF). When cutting, hold the rebar on the Stopper (HCC-13DF/16DF) or Hold Bolt (HCC-19DF) side, as shown in the figure on the left. If not, the cut piece may fly off and cause serious injury to the operator or bystanders.

3. Push in the Switch Lock on Side A. The switch is unlocked and the trigger can be operated.
4. Press the Trigger Switch to start cutting operation. The Cutter Rod will move forward to cut the rebar. Keep the Switch depressed until the Cutter Rod stops at the end of its stroke.
5. Release the Trigger Switch when the cut is completed and the Cutter Rod has reached the end of its stroke. The Cutter Rod will then return automatically to its starting position. The Cutter Rod will not return if the stroke is not completed. Similarly the Cutter Rod will not be able to move forward again until after it returns completely to its starting position. Press the switch to start the next cut, only after the Cutter Rod completely returns to its starting position and stops.



! WARNING

When cutting rebar of a high tensile strength the cut piece may fly off and cause serious injury to the operator. Wear safety glasses and confirm that the surrounding area is safe before starting operation.

! WARNING

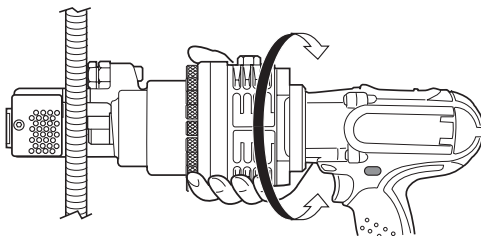


Keep your hands and face away from the blades, the moving parts and the cutting area, during operation. Remove the Battery from the tool immediately after use.



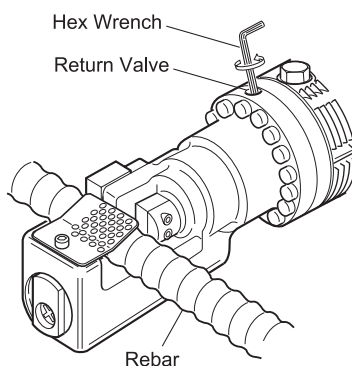
Note: Keep the air hole in the end of the Bar Holder clear of dirt and debris. The air hole controls the internal pressure and should not be obstructed.

Rotating Function of Motor (HCC-16DF & HCC-19DF)



The Motor Body can be rotated through 360 degrees, in either direction, during operation. This feature is particularly useful when working in awkward or narrow areas as it allows the operator to position the tool in the best position for easy operation.

Return Valve Operation (HCC-13DF, HCC-16DF & HCC-19DF)



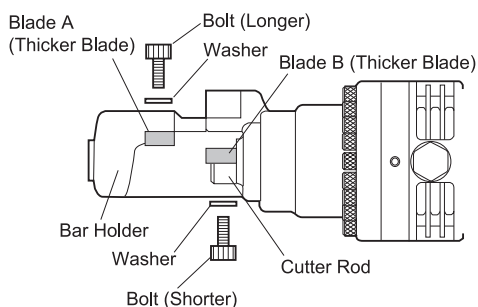
The function of the Return Valve is to allow the Cutter Rod to return to the starting position if it is unable complete a cut or becomes jammed. Using supplied hex wrench, loosen Return Valve about half a turn in anticlockwise direction. This will release the oil pressure and allow the Cutter Rod to return. Retighten the Return Valve once the Cutter Rod is fully returned and before starting the next operation.

BLADES REPLACEMENT PROCEDURE

If the cutting edges of the blades are chipped, cracked, deformed, or damaged in any way, their cutting ability will be reduced. Cutting under such conditions may cause further damage and result in personal injury. The blades should be replaced as a set immediately if any damage is found.

WARNING

When replacing the blades, ensure that the Battery is removed from the tool to prevent accidental operation.



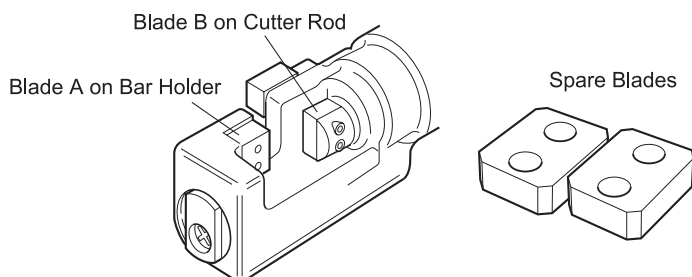
Ensure that Blade A, on the Bar Holder and Blade B, on the Cutter Rod are fitted in their correct respective positions.

1. Undo the bolts and the washers that hold Blade A and Blade B.
2. Remove dirt and clean the surfaces where the new blades are to be fitted.
3. Fit Blade A to the Bar Holder and Blade B to the Cutter Rod. Replace bolts and washers and tighten firmly.

WARNING

The bolts that hold Blade A and Blade B should be tightened regularly. If the bolts become loose the Blades may be damaged and may cause personal injury.

Type of spare blade and detachment



Securing bolts should be firmly tightened. Confirm periodically that the blade is tightened properly.

Spare blade size

※Use this table to identify the correct blades for your model.

Model	Part Number (Set)	A (Fitted to Bar Holder)	B (Fitted to Cutter Rod)
HCC-13DF (φ 3 ~ φ 13)	9002010	20 x 15 x 9 mm (Bolt Size 5 mm)	20 x 15 x 8 mm (Bolt Size 5 mm)
HCC-16DF (φ 3 ~ φ 16)	9002470	22 x 17 x 9 mm (Bolt Size 5 mm)	22 x 17 x 8 mm (Bolt Size 5 mm)
HCC-19DF (φ 3 ~ φ 20)	9002030	28 x 20 x 11.5 mm (Bolt Size 6 mm)	26 x 20 x 10 mm (Bolt Size 5 mm)

Note: Use only genuine Ogura blades.

ADDING OIL

This cordless rebar cutter is electro-hydraulic. When shipped from the factory, it was filled with oil. Do not attempt to add oil as long as the tool performs well. Over a period of time the oil level will gradually go down. Eventually this will cause a noticeable dropping off in performance. When this happens add oil as follows.

1. Place some rebar between the blades and pull the Trigger Switch.
2. Release the Trigger Switch just before the cut is completed to stop the tool.
3. Remove the Battery from the tool, so that the Blades cannot be moved accidentally.
4. Remove the Bolt (SB10X15) which caps the oil filler hole. Add the oil, being careful not to allow any oil to spill into the motor.
5. Replace the Bolt (SB10X15) and tighten securely.
6. Reinsert the Battery into the tool and complete the cutting operation.
7. Repeat the above procedure several times until the oil level remains correct.

CAUTION

Only pure hydraulic oil as recommended by Ogura & Company Ltd., should be used in this tool. Recommended oils include the Ogura supplied hydraulic oil, Super Hyrando #46 (JXTG Nippon Oil & Energy Corp.); Shell Tellus Plus #46 (U.S. Shell); or equivalent spec anti-wear hydraulic oil, ISO Viscosity Grade 46. Do not use other oils as these may cause damage to the seals and other internal machine parts.

TROUBLE SHOOTING

⚠ WARNING: Remove battery before working on machine.

PROBLEM	CAUSE	SOLUTION
Cutter Rod will not extend.	Insufficient oil	Top up oil. (Refer to “Adding Oil”)
	Cutter Rod has not returned completely due to build up of debris between Cutter Rod and Bar Holder.	Manually push back Cutter Rod. Remove debris and clean
	Cutter Rod has not returned completely due to damage to the Cutter Rod.	Replace Cutter Rod.
	Cutter Rod has not returned completely due to loose or damaged Blades.	Tighten Blade bolts. Replace Blades
	Cutter Rod has not returned completely due to weak Return spring.	Replace Return spring.
Insufficient power to cut rebar.	Insufficient oil.	Top up oil. (Refer to “Adding Oil”)
	Return Valve not properly seated or seating damaged.	Clean tip of Return Valve and Seating. Remove any scratches from seating.
	Return Valve, damaged.	Replace.
	Incorrect clearance between Cylinder and Piston.	Replace Piston (Note: different size pistons available)
	Check Valve, not properly seated or seating damaged.	Clean Check Valve and seating. Replace.
	Packing, damaged or broken.	Replace.
Oil leaks.	Oil leveller Bladder, damaged or broken.	Replace.
	Cutter Rod/Bar Holder, O-ring damaged; Cutter Rod/Bar Holder, surface scratched or grooved.	Replace Back-Up Ring and O-ring. Replace Cutter Rod/Bar Holder
	Cylinder/Bar Holder, O-ring damaged	Replace O-ring.
	Cylinder/Pump Case, Gasket damaged.	Replace liner B.
	Bar Holder/Cylinder/Pump Case, flange bolts loose.	Tighten bolts.
Motor not moving. Motor moves slow or erratic.	Voltage incorrect.	Use Proper Battery.
	Battery at end of working life.	Replace Battery.
	DC Motor damaged by over-heating.	Replace DC Motor.
	DC Motor bearings or gear damaged or broken.	Replace bearings or gear.

NOTE: The internal components of the pump and piston area have very close tolerances and are sensitive to damage from dust, dirt, contamination of the hydraulic fluid or improper handling. The disassembly of the pump housing requires special tools and training, and should only be attempted by qualified repair personnel that have been properly trained and have the right tools. The improper servicing of electrical components can lead to conditions that could cause serious injury. The pump, piston components and all electrical parts should be serviced only by authorized repair shop, dealer or distributor.

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SO.8503270/1906